

Multivariate Analysis: CA Week Five

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Question 1

We are constructing confidence intervals for:

Individual t:

$$\bar{x}_i \pm t_{n-1}(\alpha/2) \sqrt{\frac{s_{ii}}{n}}$$

Simultaneous t:

$$\bar{x}_i \pm \sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{ii}}{n}}$$

Bonferroni:

$$\bar{x}_i \pm t_{n-1}\left(\frac{\alpha}{2p}\right) \sqrt{\frac{s_{ii}}{n}}$$

Table 1: Confidence intervals for MaxBreadth

	Lower Bound	Upper Bound
Individual t	129.4514	133.2820
Simultaneous T^2	128.0909	134.6425
Bonferroni	128.8727	133.8607

Table 2: Confidence intervals for BasHeight

	Lower Bound	Upper Bound
Individual t	131.9312	135.2688
Simultaneous T^2	130.7458	136.4542
Bonferroni	131.4270	135.7730

Table 3: Confidence intervals for NasHeight

	Lower Bound	Upper Bound
Individual t	96.96939	101.3639
Simultaneous T^2	95.40858	102.9247
Bonferroni	96.30548	102.0279

Table 4: Confidence intervals for NasHeight

	Lower Bound	Upper Bound
Individual t	49.50144	51.56523
Simultaneous T^2	48.76844	52.29822
Bonferroni	49.18965	51.87702

Figure 1 compares the Individual t, Simultaneous T^2 and Bonferroni confidence intervals for the means in period 1. As expected the Individual t interval is the most narrow and the Simultaneous T^2 is the widest. The Bonferroni interval lies in between the other two. *BasLength* has the largest confidence intervals. This indicates that *BasLength* is the most variable factor in the population in time period 1. The narrowest confidence interval corresponds to *NasHeight* which provides evidence that *NasHeight* is the least variable factor in time period 1.

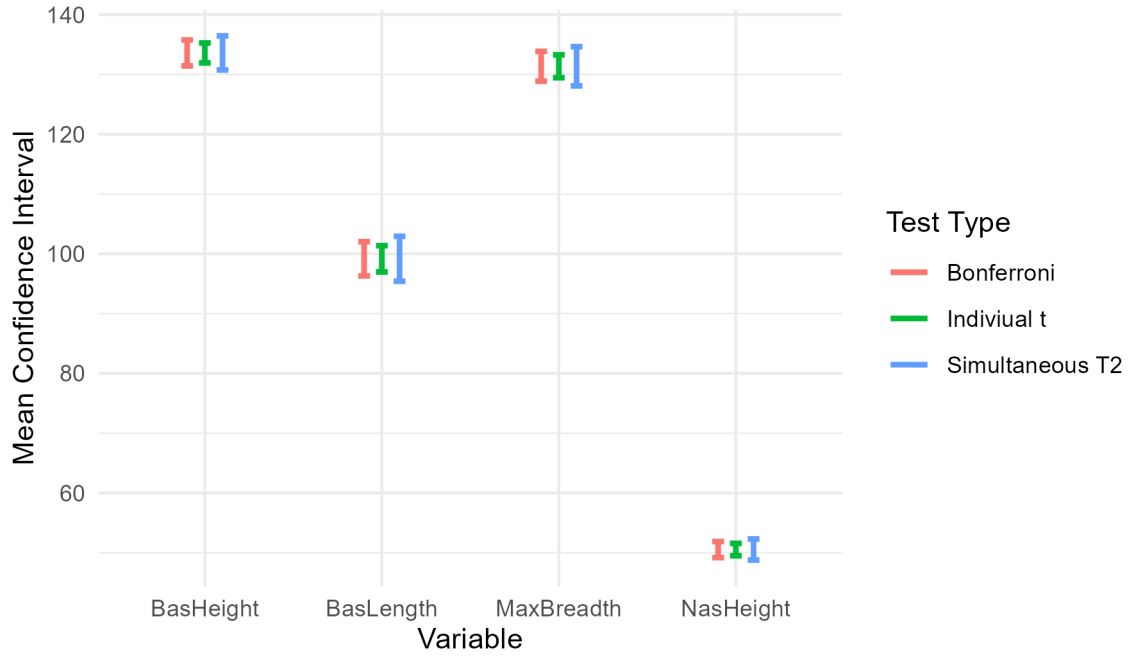


Figure 1: Comparison of Individual t, Simultaneous T^2 and Bonferroni confidence intervals for the means in period 1.

Question 2

We are testing for

$$H_o : \mu_{initial} = \mu_1 = \mu_2 = \dots = \mu_4 \quad vs \quad H_1 : \text{At least one } \mu_j \text{ differs}$$

which is equivalent to:

$$H_o : \mu_{initial} - \mu_1 = \dots = \mu_3 - \mu_4 \quad vs \quad H_1 : \text{At least one } \mu_j \text{ differs}$$

And we define our contrast matrix

$$C = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

Thus, the hypotheses are now,

$$H_o : C\mu = 0 \quad vs \quad H_1 : C\mu \neq 0$$

Where,

$$C\mu = \begin{pmatrix} \mu_{initial} - \mu_1 \\ \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \\ \mu_3 - \mu_4 \end{pmatrix}$$

The p-value is 0.01.

The very small p-value provides strong evidence that at least one time points average weight differs. The confidence intervals in Table 5 do not include 0. This suggests that there is a significant difference between time point average weights. The difference between the initial timepoint and timepoint 1, as well as the difference between timepoint 1 and timepoint 2 is positive, suggesting an increase in weight over this period. However, from timepoint 2 to 3 and from 3 to 4 the difference is negative, suggesting a decrease in weight.

The decreases in weight are of larger magnitude than the initial increase, providing evidence that after the 4 weeks the individuals has lost weight and the diet was effective.

Therefore, the resulting simultaneous T^2 confidence interval is seen in Table 5.

Table 5: Confidence intervals for repeated measures

Lower Bound	Upper Bound
0.399	0.511
1.072	1.793
-0.923	-0.717
-2.053	-0.997